

Photometric Distance Determination of a Halo RR Lyrae Variable using ASAS-SN Data and Calibrations

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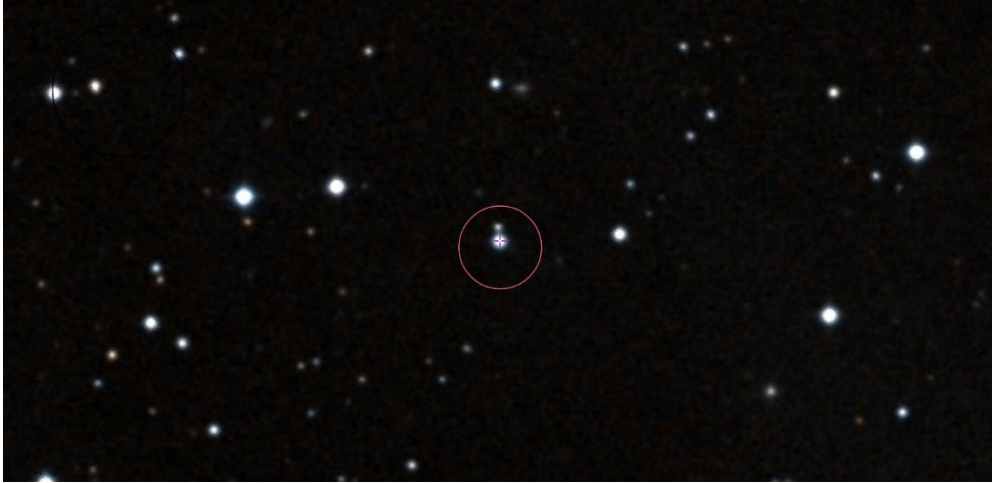


Figure 1: ASASSN-V J000023.67+361942.2 / NSVS 6313844 (0.09862, 36.32839)

image credits: ASAS-SN

Abstract

This study presents a photometric distance calculation using Python for an RR Lyrae variable star utilizing archival light curve data from the All-Sky Automated Survey for SuperNovae (ASAS-SN). By applying a Lomb-Scargle periodogram to the unevenly sampled data, I was able to identify the star's primary pulsation period. The data was subsequently phase-folded, and the Fourier series was fitted to the light curve. Using the phase parameters, I calculated the star's metallicity ($[Fe/H]$) to calibrate its absolute magnitude as a standard candle. Initial calculations revealed a significant discrepancy compared to geometric distances indicating the presence of interstellar dust. By integrating reddening data available to determine the visual extinction, I applied the distance modulus formula to correct for this cosmic fog. The final refined photometric distance of 3779.24 pc (3.78 kpc) or 12,326.21 light-years closely aligns with geometric parallax measurements, successfully placing the star within the Galactic Halo and demonstrating the critical role of extinction corrections in astronomical distance.

About RR Lyrae Stars

Measuring distances in astrophysics often relies on Standard Candles which are astronomical objects with a known, standardized intrinsic brightness (absolute magnitude). By comparing an object's true, intrinsic brightness to how faint it appears from Earth (apparent magnitude), we can precisely calculate its distance across space. RR Lyrae stars serve as crucial standard candles for mapping our galaxy. These are ancient, unstable stars typically residing in the Galactic Halo, that physically pulsate causing their brightness to rhythmically fluctuate.

Methodology and Formulae:

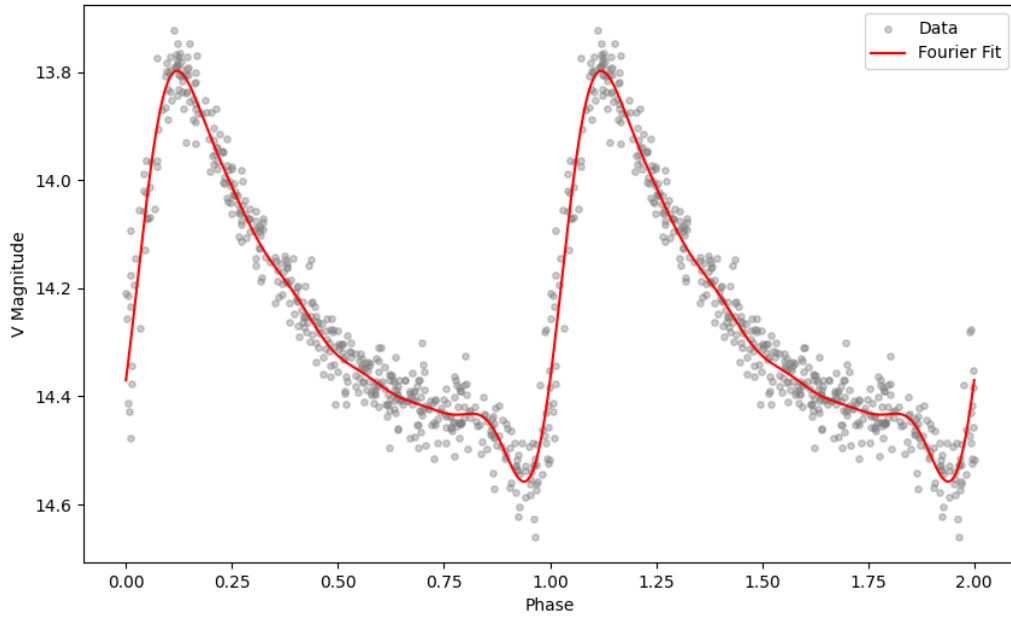


Figure 2: Phase-folded V-band light curve of the target RR Lyrae variable reproduced using the CSV file

The fluctuating brightness represents the physical pulsation of the star every 13.5 hours

Metallicity Calculation (calculating chemistry from the light curve's period and phase):

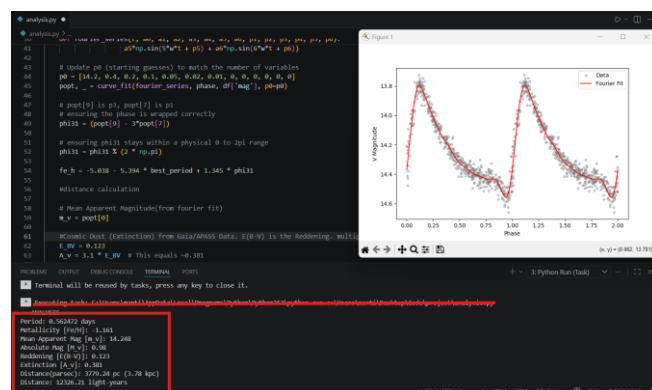
$$\left[\frac{Fe}{H}\right] = -5.038 - 5.394P + \frac{1.345}{phi_{\{31\}}}$$

Absolute Magnitude (calculating the true intrinsic brightness of the standard candle)

$$M_v = 0.59 \times \left[\frac{Fe}{H}\right] + 0.95$$

The Distance Modulus (calculating the final distance in parsec)

$$d = 10^{\left[\frac{(m_v - M_v + 5 - A_v)}{5}\right]}$$



Results

The analysis helped produce the following results:

Period	0.562472 days
Metallicity [Fe/H]	-1.161
Mean Apparent Magnitude	14.248
Absolute Magnitude	0.98
Distance (parsec)	3779.24 pc (3.78 kpc)
Distance (L.Y.)	12326.21 light-years

In this study, I successfully calculated the photometric distance to a Galactic Halo RR Lyrae star to be 3,779 parsecs which when compared to the official distance of 3,714 parsecs cataloged by the ASAS-SN observatory, yielded an error margin of 1.75%.

About this Star:

Based on the calculations, this star is an ancient, metal-poor "Population II" star, likely over 10 billion years old that formed in the early universe with only about 7% of the iron found in our Sun (since we know the metallicity). Sitting roughly 12,300 light-years away from Earth, it orbits far outside the Milky Way's familiar dusty spiral arms, high above the galactic plane in the sparse spherical cloud known as the Galactic Halo. The star is currently in its dying phase, having exhausted its core hydrogen to begin fusing helium. This structurally unstable stage causes its outer layers to act like a giant cosmic piston, physically swelling and shrinking to create the 13.5-hour (0.56-day) rhythmic pulse we observed.

link to the ASAS-SN website catalogue of The RR Lyrae Variable:

[ASAS-SN Variable Stars Database ASASSN-V J000023.67+361942.2](#)
